

MYSQL

MCQs on Databases, Tables, and Schemas

1. **Which command is used to create a new database?** a) CREATE DATABASE
b) CREATE SCHEMA
c) INSERT DATABASE
d) NEW DATABASE

Answer: a) CREATE DATABASE

2. **Which of the following is true about schemas in databases?** a) Schema is a collection of tables
b) Schema is a collection of rows
c) Schema cannot be used to organize objects in a database
d) A schema is a synonym for a database

Answer: a) Schema is a collection of tables

3. **What does the CREATE TABLE command do?** a) Deletes a table
b) Creates a new row
c) Creates a new table
d) Updates an existing table

Answer: c) Creates a new table

4. **Which clause is used to remove a database?** a) DROP TABLE
b) DELETE DATABASE
c) REMOVE DATABASE
d) DROP DATABASE

Answer: d) DROP DATABASE

5. **What is the purpose of the WHERE clause in a SQL query?** a) To define the structure of a table
b) To filter records
c) To sort records
d) To create new tables

Answer: b) To filter records

6. **Which keyword is used to filter records based on a condition in SQL?** a) WHERE
b) HAVING
c) FILTER
d) SORT

Answer: a) WHERE

7. **Which command is used to sort the results of a query in ascending order?** a) ORDER BY ASC
b) SORT BY
c) ORDER
d) ASCENDING ORDER

Answer: a) ORDER BY ASC

8. **Which SQL clause is used to group rows that have the same values?** a) GROUP BY
b) ORDER BY
c) HAVING
d) JOIN

Answer: a) GROUP BY

9. **What is the function of the HAVING clause?** a) To filter groups after the GROUP BY operation

- b) To define the structure of the table
- c) To filter individual rows
- d) To sort results

Answer: a) To filter groups after the GROUP BY operation

10. **Which SQL aggregate function is used to calculate the average value of a numeric column?** a) SUM
b) AVG
c) COUNT
d) MIN

Answer: b) AVG

11. **Which function counts the number of rows in a SQL query?** a) COUNT
b) SUM
c) AVG
d) MAX

Answer: a) COUNT

12. **Which command is used to delete a table?** a) DROP TABLE
b) DELETE TABLE
c) REMOVE TABLE
d) CLEAR TABLE

Answer: a) DROP TABLE

13. **What is the purpose of the JOIN clause in SQL?** a) To combine data from multiple tables
b) To delete data from tables
c) To create new tables
d) To modify records in a table

Answer: a) To combine data from multiple tables

14. **Which SQL JOIN returns all records when there is a match in one of the tables?**
a) INNER JOIN
b) LEFT JOIN
c) RIGHT JOIN
d) FULL OUTER JOIN

Answer: b) LEFT JOIN

15. **Which SQL JOIN returns only the records that have matching values in both tables?** a) INNER JOIN
b) LEFT JOIN
c) RIGHT JOIN
d) FULL OUTER JOIN

Answer: a) INNER JOIN

16. **What will the SQL statement `SELECT COUNT(*) FROM employees;` return?** a) The total number of columns
b) The total number of rows in the employees table
c) The number of unique rows in the employees table
d) The total sum of all employee salaries

Answer: b) The total number of rows in the employees table

17. **What does the SUM function do in SQL?** a) Returns the highest value
b) Counts the number of rows
c) Calculates the total sum of a numeric column
d) Returns the average value

Answer: c) Calculates the total sum of a numeric column

18. **What is a subquery in SQL?** a) A query within another query
b) A query that performs a join operation
c) A query that updates data
d) A query that deletes data
Answer: a) A query within another query
19. **Which type of subquery is executed once for each row processed by the parent query?** a) Inline subquery
b) Correlated subquery
c) Nested subquery
d) Scalar subquery
Answer: b) Correlated subquery
20. **Which clause is used to specify conditions for sorting data in SQL?** a) ORDER BY
b) GROUP BY
c) WHERE
d) HAVING
Answer: a) ORDER BY

Best Practices for Organizing and Naming Databases

21. **Which of the following is a good practice for naming databases?** a) Using complex names
b) Using lowercase and underscores
c) Using spaces between words
d) Using special characters in names
Answer: b) Using lowercase and underscores
22. **What should you avoid when naming tables in a database?** a) Using clear and descriptive names
b) Using a consistent naming convention
c) Using reserved keywords as table names
d) Using underscores to separate words
Answer: c) Using reserved keywords as table names
23. **Which of the following is a recommended practice when organizing a database schema?** a) Group tables based on their functionality
b) Place all tables in a single schema
c) Use arbitrary names for tables
d) Place tables with no relationship in the same schema
Answer: a) Group tables based on their functionality
24. **Which of the following should be avoided when creating databases?** a) Using descriptive names for tables
b) Creating indexes for frequently queried columns
c) Using inconsistent naming conventions
d) Normalizing the database schema
Answer: c) Using inconsistent naming conventions
25. **What is the benefit of normalizing a database?** a) Reduces redundancy and improves data integrity
b) Makes queries slower
c) Makes it easier to write complex queries
d) Increases data redundancy
Answer: a) Reduces redundancy and improves data integrity

26. **What does the DROP DATABASE command do?** a) Deletes a table from the database
b) Deletes the entire database and its contents
c) Drops an index
d) Deletes a specific column from a table
Answer: b) Deletes the entire database and its contents

SQL Syntax and Examples

27. **Which command is used to add a new column to an existing table?** a) ADD COLUMN
b) ALTER TABLE ADD COLUMN
c) ALTER COLUMN
d) CREATE COLUMN
Answer: b) ALTER TABLE ADD COLUMN
28. **What will the following query do: SELECT name FROM students WHERE age > 18 ;?** a) Retrieve all students' names
b) Retrieve the names of students who are older than 18
c) Retrieve the age of students who are older than 18
d) Retrieve all students' names who are younger than 18
Answer: b) Retrieve the names of students who are older than 18
29. **How do you select all columns from the "employees" table?** a) SELECT * FROM employees
b) SELECT all FROM employees
c) SELECT columns FROM employees
d) SELECT ALL columns FROM employees
Answer: a) SELECT * FROM employees
30. **Which of the following SQL queries is used to fetch all records from the "customers" table where the city is "New York"?** a) SELECT * FROM customers WHERE city = 'New York';
b) SELECT city FROM customers WHERE city = 'New York';
c) SELECT * FROM customers HAVING city = 'New York';
d) SELECT FROM customers WHERE city = 'New York';
Answer: a) SELECT * FROM customers WHERE city = 'New York';
31. **Which SQL clause is used to filter data after grouping?** a) WHERE
b) HAVING
c) GROUP BY
d) ORDER BY
Answer: b) HAVING
32. **What does the DISTINCT keyword do in a SQL query?** a) Retrieves duplicate values
b) Retrieves unique values
c) Sorts the results
d) Limits the number of rows
Answer: b) Retrieves unique values
33. **Which SQL statement is used to remove all rows from a table without deleting the table itself?** a) DROP TABLE
b)

DELETE TABLE
c) TRUNCATE TABLE

d) REMOVE TABLE

Answer: c) TRUNCATE TABLE

34. Which command is used to update data in an existing record? a) MODIFY

b) UPDATE

c) ALTER

d) SET

Answer: b) UPDATE

35. Which SQL clause is used to specify the condition for filtering records based on an aggregate function? a) WHERE

b) HAVING

c) GROUP BY

d) ORDER BY

Answer: b) HAVING

36. Which operator is used to check for NULL values in SQL? a) IS NULL

b) IS EMPTY

c) NULL

d) NOT NULL

Answer: a) IS NULL

37. Which of the following SQL statements is correct for creating a table with three columns (ID, Name, Age)? a) CREATE TABLE users (ID INT, Name VARCHAR, Age INT);

b) CREATE TABLE users (ID INT, Name STRING, Age INT);

c) CREATE TABLE users (ID INTEGER, Name VARCHAR, Age INTEGER);

d) CREATE TABLE users (ID, Name, Age);

Answer: c) CREATE TABLE users (ID INTEGER, Name VARCHAR, Age INTEGER);

38. Which operator is used to match a pattern in SQL? a) LIKE

b) MATCH

c) PATTERN

d) SEARCH

Answer: a) LIKE

39. What is the result of the SQL query `SELECT * FROM employees WHERE salary BETWEEN 50000 AND 70000;`? a) Selects employees with a salary less than 50,000

b) Selects employees with a salary between 50,000 and 70,000

c) Selects employees with a salary more than 70,000

d) Selects employees with a salary greater than or equal to 70,000

Answer: b) Selects employees with a salary between 50,000 and 70,000

40. Which of the following SQL commands is used to insert a new row into a table?

a) INSERT INTO

b) ADD ROW

c) INSERT NEW

d) CREATE ROW

Answer: a) INSERT INTO

Advanced Queries

41. Which SQL function returns the number of rows in a table or a query? a)

COUNT

b) SUM

- c) AVG
- d) MAX

Answer: a) COUNT

42. **Which command is used to create a new database schema?** a) CREATE SCHEMA
b) CREATE DATABASE SCHEMA
c) CREATE DATABASE
d) NEW SCHEMA

Answer: a) CREATE SCHEMA

43. **What is the correct syntax for using a subquery in the WHERE clause?** a) SELECT * FROM employees WHERE salary = (SELECT MAX(salary) FROM employees);
b) SELECT * FROM employees WHERE salary = [SELECT MAX(salary) FROM employees];
c) SELECT * FROM employees WHERE salary = <SELECT MAX(salary) FROM employees>;
d) SELECT * FROM employees WHERE salary = SELECT MAX(salary) FROM employees;

Answer: a) SELECT * FROM employees WHERE salary = (SELECT MAX(salary) FROM employees);

44. **Which SQL statement retrieves all employees whose salary is greater than the average salary?** a) SELECT * FROM employees WHERE salary > AVG(salary);
b) SELECT * FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);
c) SELECT * FROM employees WHERE salary > AVG(salary) GROUP BY salary;
d) SELECT * FROM employees WHERE salary > MAX(salary);

Answer: b) SELECT * FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);

45. **Which keyword is used to sort the result set of a query in descending order?** a) DESCENDING
b) DESC
c) ORDER DESC
d) SORT DESC

Answer: b) DESC

46. **Which clause is used to specify the column or columns to join in an SQL JOIN operation?** a) ON
b) USING
c) JOIN
d) WHERE

Answer: a) ON

47. **Which clause is used to filter the results of a query based on an aggregate function?** a) HAVING
b) WHERE
c) GROUP BY
d) SELECT

Answer: a) HAVING

48. **Which SQL statement would be used to change an existing record in a table?** a) MODIFY
b) ALTER
c) UPDATE

d) CHANGE

Answer: c) UPDATE

49. Which function is used to find the largest value in a column? a) MAX

b) MIN

c) COUNT

d) AVG

Answer: a) MAX

50. Which of the following SQL queries is correct for joining two tables **orders** and **customers** on a common column **customer_id**? a) SELECT * FROM orders JOIN customers ON orders.customer_id = customers.customer_id;

b) SELECT * FROM orders INNER JOIN customers WHERE orders.customer_id = customers.customer_id;

c) SELECT * FROM orders FULL JOIN customers ON orders.customer_id = customers.customer_id;

d) SELECT * FROM orders LEFT JOIN customers WHERE orders.customer_id = customers.customer_id;

Answer: a) SELECT * FROM orders JOIN customers ON orders.customer_id = customers.customer_id;